

**Time:** 5 minutes**Outcome:** Apply open-market operations, administered rates, and reserve tools.**Difficulty:**

Intermediate

Monetary-Policy Tools



THE CORE IDEA

The Federal Reserve influences bank reserves, the money supply, and short-term interest rates through several policy tools. Open-market purchases add reserves to the banking system, while open-market sales remove them. Administered rates change the incentives banks face when borrowing or holding reserves. Changes in reserves can affect bank lending and deposit creation, although the actual expansion of money may be smaller than the simple textbook multiplier predicts.



HOW TO RECOGNIZE IT

- An open-market purchase adds reserves and tends to lower short-term interest rates.
- An open-market sale removes reserves and tends to raise short-term interest rates.
- Lower administered rates encourage lending and monetary expansion; higher rates restrain them.
- Trace the direction of reserves, bank lending, money, and interest rates for each policy tool.



WATCH OUT

Do not assume every tool in a policy package pushes bank lending in the same direction. The open-market purchase is expansionary, while a higher return on reserves can weaken the lending response.



WORKED EXAMPLE: MONEY MULTIPLIER IN ACTION

The central bank buys \$50 million of securities, adding \$50 million to bank reserves. If banks lend 80% of new reserves and the deposit multiplier is 5, the effective multiplier is $0.80 \times 5 = 4$. The potential increase in deposits is therefore $\$50 \text{ million} \times 4 = \200 million , smaller than the \$250 million simple-model maximum.



CHECK YOURSELF

If the central bank lowers the rate it charges banks for direct loans, what is the immediate textbook effect?

**READY?**

Return to the game and master this concept.